



The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	FMA4800
Course title (English)	Financial Computation
Course title (Chinese)	金融计算
Units	3
Description (English)	This course introduces several important numerical techniques (numerical PDE, Monte Carlo simulation, etc.) for computational problems in finance including derivatives pricing, risk management and portfolio optimization.
Description (Chinese)	本课程主要介绍一些重要的数值方法（例如数值偏微分方程、蒙特卡洛方法等）来解决金融中的计算问题，包括衍生品定价、风险定价以及投资组合优化。

B. Corresponding course at CUHK

Please give details of the *closest* corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	SEEM4720
Course title (English)	Computational Finance
Course title (Chinese)	计算金融学
Units	3
Description (English)	The course of Computational Finance provides a quantitative introduction to derivative markets from the computational perspective. In it, we will focus on (i) the fundamental mechanics of futures, swaps and option markets, (ii) risk neutral evaluation theory of asset pricing, (iii) numerical procedures related to derivative evaluation and risk managements. Using MATLAB as a means of illustration, the course focuses on developing numerical problem-solving capabilities for the students.
Description (Chinese)	本课程从计算的角度介绍金融市场，包括期货、互换、期权



	市场的基本机制，风险中性定价理论，衍生品定价和风险管理的数值方法。本课程使用MATLAB来增强学生应用数值方法解决实际问题的能力。
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2. Course requisites

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or any other requirements.

A. Prerequisites

Having taken courses linear algebra, calculus, probability theory, stochastic calculus and stochastic differential equations.

B. Co-requisites

C. Others

3. Learning outcomes

Knowledge:

- a) Students will understand the mechanics of financial markets and the important computational finance problems.
- b) Student will understand stochastic models of finance.
- c) Students will understand finite difference method and its applications in solving finance problems.
- d) Students will understand Monte Carlo method and its applications in solving finance problems.

Skills

Generic:

- e) Students will be able to solve derivatives pricing problems with stochastic models and numerical methods.
- f) Students will be able to evaluate and manage financial risk with stochastic models.

Valued/Attitude:

- g) Students will have awareness of the power of stochastic models and numerical methods in solving real-world problems.

4. Course syllabus

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.



Introduction to financial markets and computational problems.
Stochastic modelling for finance.
Monte Carlo simulation.
Binomial tree.

5. Assessment scheme

Component/ method	% weight
Assignments	30
Mid-term exam	30
Final exam	40

6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.

Type: **General**

Grade	Description
A	Demonstrates the ability to achieve all the learning outcomes in this course and be able to apply the principles in solving problems in novel situations, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials



	Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.

7. Feedback for evaluation

Please specify forms of evaluation to generate feedback on classes from students.

A course questionnaire (CTE) was answered by each student after completing all teaching activities.

Reflections from teachers of more advanced courses.

Feedback from informal conversations over in the instructor's office and wechat/e-mail correspondence with students after class.

8. Readings

9. Course components

Activity	Hours/week
Lectures	3
Readings & Assignments	6
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
00000 00001	Introduction to financial markets and computational problems.
00000 00002	Stochastic modelling for finance.
00000 00003	Monte Carlo simulation.
00000 00004	Monte Carlo simulation.
00000 00005	Monte Carlo simulation.



00000 00006	Monte Carlo simulation.
00000 00007	Monte Carlo simulation.
00000 00008	Monte Carlo simulation.
00000 00009	Monte Carlo simulation.
00000 00010	Binomial tree.
00000 00011	Binomial tree.
00000 00012	Binomial tree.
00000 00013	Binomial tree.
00000 00014	Binomial tree.

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

Expected enrolment: 50

Number of lecturing sections: 13

Number of tutorial sections: 10-12

12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before using AI in academic work, and be sure to acknowledge any AI use in submissions.

13. Approval



Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

14. Any other information

15. Version date

Version number	4
As of (date)	2025/12/02